



Semester 1 Examinations 2018 / 2019

Exam Code(s) 4BCT1, 4BSE1, 1MSME1, 1MEEE1, 1MECS1, 1CSD1, 1CSD2, 2SPE1

Exam(s) B.Sc. Degree (Computer Science and Information Technology)
B.E. Energy Systems Engineering
MSc in Mechanical Engineering
ME (Electrical and Computer Engineering)
ME (Electrical and Electronic Engineering)
M.Sc. in Computer Science (Data Analytics)

Module Code(s) CT561

Module(s) Systems Modelling and Simulation

Paper No. I

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Internal Examiner(s) Prof. Michael Madden
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Instructions:

Answer any 3 questions. All questions carry equal marks.

Duration 2hrs

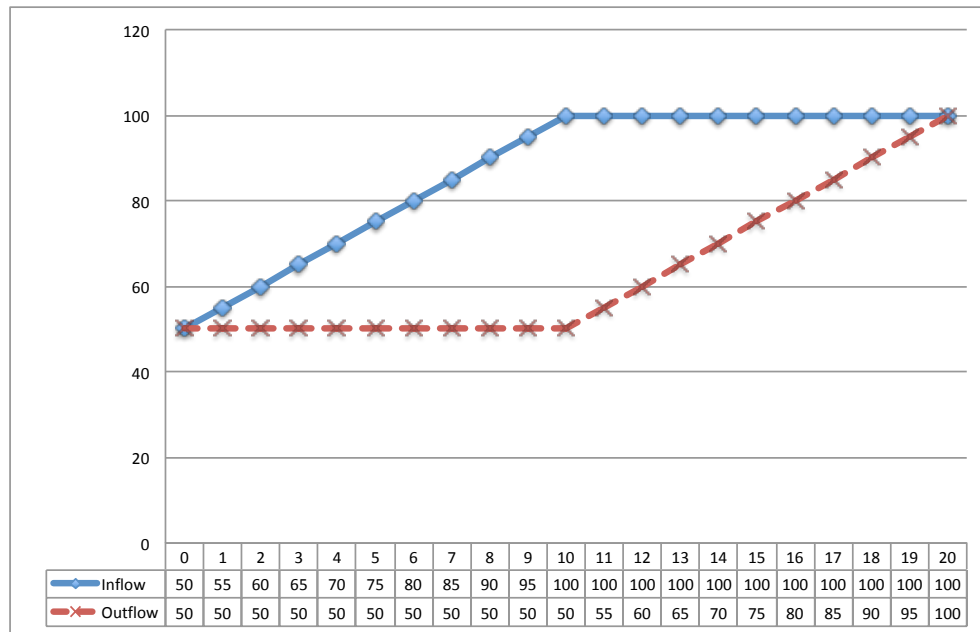
No. of Pages 5 (Including Cover Page)

Department(s) Information Technology

Requirements Graph Paper

1. (a) Based on the flow variables shown in the diagram, and assuming that the initial value of the stock is 100, perform the following:

- Calculate, and show, the net flow.
- Use graphical integration to plot the stock over time, clearly showing the behaviour mode for each segment (e.g. *increasing at an increasing rate, etc.*)



[10]

- (b) Calculate the stock using Euler's equation where $DT=5$. Estimate the integration error for each time step of 5.

[8]

- (c) Draw a stock and flow model of the variables. Show how Little's Law can be used to show that the steady state delay value (from time 20 onwards) will be 6. What kind of delay distribution is suggested by the behaviour of the outflow when compared to that of the inflow?

[7]

2. (a) Construct a stock and flow model from the following description of a Health Care system. Specify the stock equation (there is no need to document equations for flows or auxiliaries). Clearly show the link polarities on the diagram.

- Patients are **scheduled** for treatment, and this increases the number of **patients waiting**. An increase in **treatments completed**, causes a decrease in **patients waiting**.
- As the number of **patients waiting** increases, so too does the **health care system pressure**.
- An increase in **health care system pressure** leads to two system responses:
 - The **time per procedure** is reduced.
 - The **workweek** is increased.
- An increase in **time per procedure** leads to a decrease in **treatments completed**.
- A decrease in **time per procedure** leads to an increase in **follow-up visits**, which in turn, will increase **patients waiting**.
- An increase in **workweek** leads to an increase in **treatments completed**.
- An increase in **workweek** leads to an increase in **fatigue** (after a delay), and increasing **fatigue** leads to an increase in **time per procedure**.

[8]

- (b) Based on the stock and flow model in part(a):

- Show the following feedback loops on the model: *"Midnight Oil"*, *"Burnout"*, *"Corner Cutting"*, and *"Haste Makes Waste"*.
- Starting with the proposition that the stock (**patients waiting**) is increasing, calculate the polarity for each of these loops, and comment on how the loops would impact the stock over time.

[12]

- (c) What are the advantages of building this feedback model? How can its insights be use to identify ways to improve the Health Care System?

[5]

3. (a) Given a boarding school with 500 students ($N=500$), calculate the force of infection if 10 students are infectious, and 480 are susceptible. Assume $R_0 = 4$, and the recovery delay $D = 2$.

Given this scenario, show why an epidemic would spread through the school.

[5]

- (b) Show the Stocks, Flows and Feedbacks for the SIR Model.

Show the structure and polarity of the following loops: *"Contagion"*, *"Depletion"* and *"Recovery"*.

Explain the formulation of the number of infections (inflow to Infectious). Also, answer the following:

- What are the plausible range of values for the parameters of the equation?
- How do the parameters of the SIR model relate to the reproduction number?
- If the model was to be extended to include people wearing personal protective equipment (PPE), for example, face masks during an influenza outbreak, what parameter could be modified? How might PPE resources be added to the SIR model?

[12]

- (c) Show that when an outbreak is at its peak ($dI/dt = 0$), the proportion susceptible equals $1/R_0$.

Explain why the SIR model conforms to the following statement: "The chain of transmission eventually breaks down due to a decline in infectives, not due to a complete lack of susceptibles."

[6]

4. (a) Describe the key idea underlying the stock management structure, and show how the *anchor* and the *adjustment* elements of the model are formulated.

[5]

- (b) Create a stock and flow model (with full equations) for the following description. It should contain a structure for modeling the effect of the **student staff ratio** on the **student attrition fraction**, and also model staff recruitment as a *stock management structure*.

- **Students** are increased by **enrollments**, and decreased by **graduations** and **attritions**.
- All inflows & outflows to students are governed by fractional increase/decrease rates.
- The initial value of students is 10000. The enrollment fraction for students is 25%, and the graduation fraction is 24%. The **attrition fraction** depends on the student to staff ratio.
- The normal attrition fraction is 1%, which applies when the **student staff ratio** is 20. If the student staff ratio increases over 20, then the attrition fraction should increase (up to a maximum of 10%). Should the student staff ratio fall below 20, then the attrition fraction should decline to a minimum of 0.1%. Show the appropriate effect function on a graph.
- **Staff** are increased by **hires** and reduced by **retirements** and **turnover**.
- The **desired student to staff ratio** is 20. This then determines the **desired staff number**.
- The inflow to staff (hires) is governed by the stock management structure. The **adjustment time** is 1. **Expected losses** are initially 10/year, where the retirement fraction is 8% and the turnover fraction is 2%. The **expectation adjustment time** is 3.

[12]

- (c) Consider the following scenario. The model starts running in 2015. Assume it runs in equilibrium. After three years, the **desired student to staff ratio** is reduced to 10. Assume the model will then run until 2030.

Show your understanding of how the system will respond by sketching the responses of following variables (for simplification, assume that the number of students will remain constant). **Student attrition fraction**, **Student to staff ratio**, **Number of staff** and **Expected staff losses**.

[8]